

MINI PROJECT IN C++

[SECURED DEPARTMENTAL STORE MANAGEMENT]

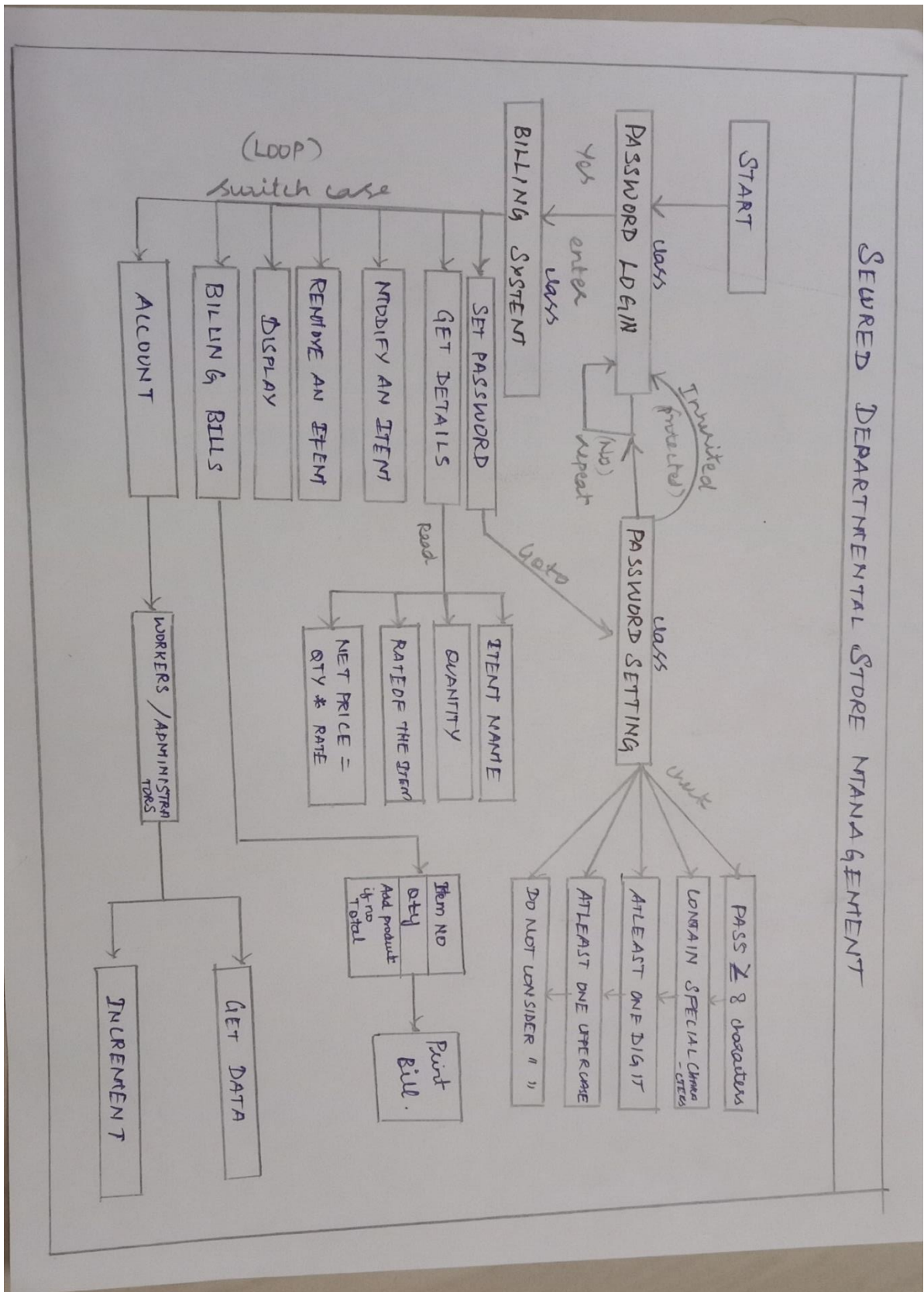
OBJECTIVE :

The project deals with the management of “AUTOMATION OF SUPERMARKET”. It is a place where entire goods are available. The objective is to automate its existing manual system by the help of computerized equipment and full fledged computer program , fulfilling their requirements ,so that their valuable data can be stored for a long period with easy accessing and manipulation of the same assist the staff in capturing the effort spent on their respective working areas.

To utilize human resource of the store in a effective manner by increasing their efficiency through automation

The system generates a number of types of reports that can be then used for various purpose

ARCHITECTURE :



PROGRAM CODE :

```
#include <iostream>
#include <string.h>
#include <stdlib.h>
#include <ctype.h>
#include <iomanip>
char a=' ';
using namespace std;
class billing
{
    private:
        char name[20];
        int qty;
        float price, net;
    public:

        void getdata();
        void modify(billing *o, int no);
        void Remove(billing *o, int no);
        void Display() ;

};

void billing :: getdata()
{
    cout<<"\n Enter Item Name      : ";
    cin>>name;
    cout<<" Enter Item Quantity (in kg): ";
    cin>>qty;
    cout<<" Enter Item Price (in Rs) : ";
    cin>>price;
    net = price * qty;
}

void billing :: modify(billing *o, int no)
{
    char key[15];
    int i,j;
    cout<<"\n Enter the Item Name to modify: ";
    cin>>key;

    for(i=0;i<=no;i++)
    {
        if(strcmp(o[i].name, key) == 0)
```

```

    {
        cout<<"\n  Enter Item Name   : ";
        cin>>o[i].name;
        cout<<"\n  Enter Item Price   : ";
        cin>>o[i].price;
        cout<<"\n  Enter Item Quantity : ";
        cin>>o[i].qty;
        o[i].net = o[i].price * o[i].qty;
        cout<<"\n\t##### RECORD MODIFIED SUCCESSFULLY !
##### ";

        break;
    }
    else if(strcmp(o[i].name, key) != 0)

        cout<<"\n\t##### ITEM DOES NOT MATCH
!!##### ";

}

}

```

void billing :: Remove(billing *o, int no)

```

{

    char key[15];
    int i,j,flag;
    cout<<"\n  Enter the Item Name to Remove: ";
    cin>>key;

    for( i=0; i<=no; i++)
    {
        if(strcmp(o[i].name, key) == 0)
        {
            for(j=i; j<=no; j++)
                o[j] = o[j+1];
            no--;
            cout<<"\n\t##### RECORD REMOVED SUCCESSFULLY !
##### ";

            break;
        }
        else if(strcmp(o[i].name, key) != 0)

            cout<<"\n\t##### ITEM DOES NOT MATCH
!!##### ";
    }
}

```

```
}
```

```
}
```

```
void billing :: Display()
```

```
{
```

```
    cout<<"\n\t Item Name: "<<name;
```

```
    cout<<"\n\t Item Price: "<<price;
```

```
    cout<<"\n\t Item Quantity: "<<qty;
```

```
    cout<<"\n\t Net Price: "<<net;
```

```
}
```

```
class passwordset
```

```
{
```

```
private :
```

```
    int i,length,count=0,ct=0,cnt=0;
```

```
    int flag1=1,flag2,flag3,flag4,flag5;
```

```
    char ps[15];
```

```
    int ch=1;
```

```
public:
```

```
    passwordset()
```

```
{
```

```
        count=0,ct=0,cnt=0;
```

```
}
```

```
    string newpasswordy();
```

```
};
```

```
string passwordset::newpasswordy()
```

```
{
```

```
while(flag1!=0 || flag2!=0 || flag3!=0 || flag4!=0 || flag5!=0)
```

```
{
```

```
    cout<<"\nENTER NEW PASSWORD : ";
```

```
    cin>>ps;
```

```

length=strlen(ps);
count=0;cnt=0;ct=0;

if(strlen(ps)<8)
{
    cout<<"YOUR PASSWORD SHOULD BE ATLEAST 8 CHARACTERS LONG";
}
else

    flag1=0;

for(i=0;i<length;i++)
{

    if(ps[i]==' ')
    {
        cout<<"YOUR PS SHOULD NOT CONTAIN BLANK SPACE\n";
    }

    else

        flag2=0;

    if(ps[i]!='&'&&ps[i]!='*'&&ps[i]!='$'&&ps[i]!='#'&&ps[i]!='@')
    {
        count++;
        if(count==length)
            cout<<"\nYOUR PASSWORD SHOULD CONTAIN ATLEAST ONE SPECIAL CHARACTER\n";

    }
    else
        flag3=0;

    if(isupper(ps[i])==0)
    {
        ct++;
        if(ct==length)

            cout<<"YOUR PASSWORD SHOULD CONTAIN ATLEAST ONE UPPER CASE CHARACTER\n";
    }
    else
        flag4=0;

    if(isdigit(ps[i])==0)
    {

        cnt++;
        if(cnt==length)
        {

```

```

        cout<<"YOUR PASSWORD MUST CONTAIN ATLEAST ONE NUMBER\n";
    }

    else
        flag5=0;

}
}
}

if(flag1==0&&flag2==0&&flag3==0&&flag4==0&&flag5==0)
{
    cout<<"\n*STRONG PASSWORD*";
    cout<<"\t\t.PASSWORD SET SUCCESSFULLY.\n\n";
}
return ps;
}

class passlogin :protected passwordset
{
public:

    void password(string r)
    {
        string opass=r,epass;

        cout<<endl;
        cout<<"\t\t\tWLCOME TO SECURED GROCERRY SHOP STACK MANAGEMENT SYSTEM \n";
        cout<<"\t\t\t_____ \n";
        cout<<"\n\n ENTER YOUR PASSWORD TO LOGIN : " ;
        cin>>epass;
        if(epass==opass)
        {
            cout<<" \t\t PASSWORD IS CORRECT \n ";

        }
        else
        {
            cout<<"\n\t\t OOPS !! ENTERED PASSSSWORD IS WRONG ! TRY AGAIN !\n\n";
            password(r);
        }
    }

};

int main()

```

```
{  
    billing b[20], bl;  
    passwordset p;  
    passlogin p2;  
  
    int ch1, item_no = -1,i;  
    string r="hello";  
    char ch2;  
  
    p2.password(r);  
  
    while(1)  
  
        {   cout<<endl;  
            cout<<setw(30);  
            cout<<"\n\n\t\t\t\t\t GROCERRY SHOP\n";  
            cout<<"\t\t\t\t\t_____ \n";  
  
            cout<<"\n\t\t\t\t\t1. Get Item Details.";  
            cout<<"\n\t\t\t\t\t2. Display Items.";  
            cout<<"\n\t\t\t\t\t3. modify an Item.";  
            cout<<"\n\t\t\t\t\t4. Remove an Item.";  
            cout<<"\n\t\t\t\t\t5. Set new password.";  
            cout<<"\n\t\t\t\t\t6. Exit. \n";  
            cout<<"\t\t\t\t\t_____ \n";  
  
            cout<<"\n Enter your choice: ";  
            cin>>ch1;  
            switch(ch1){  
                case 1:  
                    while(1)  
  
                        {  
  
                            item_no++;  
  
                            cout<<"\n Entry of Purchased Items.";  
                            cout<<"\n -----";  
                            cout<<endl;  
                            cout<<"\n Item No. "<<item_no+1;  
                            b[item_no].getdata();  
                            cout<<"\n Do you want to add more(y/n): ";  
                            cin>>ch2;  
  
                            if(ch2 == 'n')  
                                break;  
  
                        }  

```



```
break;
```

case 2:

```
cout<<"\n\n Purchase order Details.";
cout<<"\n -----";
for(int i=0; i<=item_no; i++)
{
    cout<<"\n\n Item No.: "<<i+1;
    b[i].Display();
}
break;
```

case 3:

```
bl.modify(b, item_no);
break;
```

case 4:

```
bl.Remove(b, item_no);
break;
```

case 5:

```
r= p.newpasswordy();
p2.password(r);
break;
```

default:

```
    exit(1);
}
```

```
}
```

DESCRIPTION WITH ITS OUTPUT :

Login to the program :

passwordlogin() Enter the password to enter into the program
.if password is incorrect retry the function until enters correct password

```
WLCOME TO SECURED GROCERRY SHOP STACK MANAGEMENT SYSTEM
_____

ENTER YOUR PASSWORD TO LOGIN : 123

OOPS !! ENTERED PASSSSWORD IS WRONG ! TRY AGAIN !

WLCOME TO SECURED GROCERRY SHOP STACK MANAGEMENT SYSTEM
_____

ENTER YOUR PASSWORD TO LOGIN : hello
PASSWORD IS CORRECT

GROCERRY SHOP
_____

1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
_____
```

1. Get item details :

`getdata()` - getting item number,item name ,item price, quantity of the item ,net price and storing this details in a object.

```
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
```

Enter your choice: 1

Entry of Purchased Items.

Item No. 1

```
Enter Item Name      : soap
Enter Item Quantity (in kg): 1
Enter Item Price (in Rs)  : 34
```

Do you want to add more(y/n): y

Entry of Purchased Items.

Item No. 2

```
Enter Item Name      : paste
Enter Item Quantity (in kg): 1
Enter Item Price (in Rs)  : 40
```

Do you want to add more(y/n): y

Entry of Purchased Items.

Item No. 3

```
Enter Item Name      : brush
Enter Item Quantity (in kg): 1
Enter Item Price (in Rs)  : 20
```

Do you want to add more(y/n): y

Entry of Purchased Items.

Item No. 4

```
Enter Item Name      : shampoo
Enter Item Quantity (in kg): 1
Enter Item Price (in Rs)  : 5
```

Do you want to add more(y/n): n

2.Display :

display() - printing the item details stored.

```
GROCERRY SHOP
-----
1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
-----

Enter your choice: 2

Purchase order Details.
-----

Item No.: 1
    Item Name:  soap
    Item Price: 34
    Item Quantity: 1
    Net Price: 34

Item No.: 2
    Item Name:  paste
    Item Price: 40
    Item Quantity: 1
    Net Price: 40

Item No.: 3
    Item Name:  brush
    Item Price: 20
    Item Quantity: 1
    Net Price: 20

Item No.: 4
    Item Name:  sampoo
    Item Price: 5
    Item Quantity: 1
    Net Price: 5
```

3.Modify an item :

`modify()` — modify an item by entering the key (item name)
if it is there it allows to change the name or quantity or price of the item .
else it does not modify any item as there is no item in the list.

```
GROCERRY SHOP
-----
1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
-----

Enter your choice: 3

Enter the Item Name to modify: oil

##### ITEM DOES NOT MATCH !!#####

GROCERRY SHOP
-----
1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
-----

Enter your choice: 3

Enter the Item Name to modify: soap

Enter Item Name      : soap

Enter Item Price      : 1

Enter Item Quantity   : 45

##### RECORD MODIFIED SUCCESSFULLY ! #####
```

4.Remove an item :

`remove()` — remove an item by entering the key(item name)

if it is there it deletes the item's details and continues order of the item number .else it does not remove any item from the list .

```
GROCERRY SHOP
-----
1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
-----

Enter your choice: 4

Enter the Item Name to Remove: oil

##### ITEM DOES NOT MATCH !!#####

GROCERRY SHOP
-----
1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
-----

Enter your choice: 4

Enter the Item Name to Remove: soap

##### RECORD REMOVED SUCCESSFULLY ! #####
```

3.Set new password :

Set password()- read the new password and checks the following conditions

- Password is greater than 8 characters .
- Password contains atleast one special character.
- Password contains atleast one digit.
- Password contains atleast one upper case character .

If all the conditions satisfies it set the new password and enter into login password .

```
1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.

Enter your choice: 5

ENTER NEW PASSWORD : sara
YOUR PASSWORD SHOULD BE ATLEAST 8 CHARACTERS LONG
YOUR PASSWORD SHOULD CONTAIN ATLEAST ONE SPECIAL CHARACTER
YOUR PASSWORD SHOULD CONTAIN ATLEAST ONE UPPER CASE CHARACTER
YOUR PASSWORD MUST CONTAIN ATLEAST ONE NUMBER

ENTER NEW PASSWORD : saravana12@SK
*STRONG PASSWORD* .PASSWORD SET SUCCESSFULLY.

WLCOME TO SECURED GROCERRY SHOP STACK MANAGEMENT SYSTEM

ENTER YOUR PASSWORD TO LOGIN : saravana
OOPS !! ENTERED PASSSSWORD IS WRONG ! TRY AGAIN !

WLCOME TO SECURED GROCERRY SHOP STACK MANAGEMENT SYSTEM

ENTER YOUR PASSWORD TO LOGIN : saravana12@SK
PASSWORD IS CORRECT

GROCERRY SHOP

1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
```

3.EXIT :

Exit(0) - exit the program

```
GROCERRY SHOP
-----
1. Get Item Details.
2. Display Items.
3. modify an Item.
4. Remove an Item.
5. Set new password.
6. Exit.
-----

Enter your choice: 6

Process returned 1 (0x1)   execution time : 76.230 s
Press any key to continue.
```

3.RESULT :

The above program is successfully executed and output is verified .

SUBMITTED BY :

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